

# NAGESHWAR PAWAR

Software Engineer | Full Stack Development (React, TypeScript, Node.js) | Microservices & AI Integration

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## PROFESSIONAL SUMMARY

Software Engineer with 1.5+ years of professional experience specializing in Full Stack Development, combining strong frontend expertise with production-grade backend architecture. Proficient in building scalable, type-safe applications with React, TypeScript, Node.js, Express, and MongoDB, with hands-on experience designing microservices-based systems, reusable shared packages, and secure authentication architectures. Experienced in integrating AI and Computer Vision capabilities into production applications, including real-time virtual try-on experiences built with TensorFlow.js and MediaPipe. Skilled in performance optimization, clean architecture, and building maintainable production systems used by real users.

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, HTML5, CSS3

**Frontend:** React.js, Next.js, Redux Toolkit, React Query, Zustand, React Hook Form, Tailwind CSS, Material UI

**Backend:** Node.js, Express.js, REST APIs, JWT, OAuth, RBAC, Multer, Zod, Cron Jobs

**Databases & Cloud:** MongoDB, Redis, Cloudinary, Render, Vercel, Netlify

**Architecture:** Microservices, MVC, Monorepo, API Gateway, Caching, Queue Systems (BullMQ), Logging, Event-driven Architecture

**AI / Computer Vision:** TensorFlow.js, MediaPipe, FaceMesh, Pose Detection, Hair Segmentation, Canvas API, Camera APIs, Image Processing

**Tools:** Git, GitHub, Bitbucket, Postman, VS Code, npm

## PROFESSIONAL EXPERIENCE

### Frontend SDE I — CTRUH

Oct 2024 – Mar 2026

On-site, Bengaluru, India | AI infrastructure company building the Spatial Internet: 3D, XR, and AI-powered immersive experiences via a low-code/no-code platform

- Developed and shipped 100+ production React.js/TypeScript pages and 100+ reusable UI components, converting Figma designs into pixel-perfect, responsive interfaces.
- Integrated 100+ REST APIs across frontend features, collaborating daily with backend, AI/ML, design, and QA teams within an 8-engineer squad.
- Built and integrated AI/Computer Vision-powered virtual try-on features (Hair Color, Makeup, Clothing, Jewellery) and Clothing Measurement using TensorFlow.js, MediaPipe, FaceMesh, Pose Detection, and Hair Segmentation.
- Engineered real-time, camera-based rendering pipelines using Canvas API, RequestAnimationFrame, image capture/processing, and mask overlay techniques for live AR experiences.
- Optimized component rendering performance by reducing unnecessary re-renders and implementing caching strategies, improving overall application responsiveness.
- Diagnosed and resolved 20+ production issues, ensuring platform stability and reliability across releases.
- Drove technical quality through regular PR/code reviews, architecture discussions, task estimation, and client-facing technical discussions.
- Collaborated with AI/ML engineers to integrate Google MediaPipe models and Google Gemini API into production workflows where applicable.
- Used Git, GitHub, and Bitbucket for version control and collaborative development across a distributed engineering team.

**Growing Star Award** — Recognized for exceptional ownership and commitment by successfully delivering a high-priority project within strict deadlines while maintaining quality and reliability.

## FLAGSHIP PROJECT

### Beautinique

Actively Developing

Production-grade Full Stack MERN E-Commerce Platform | Microservices Architecture with API Gateway & Monorepo

Scale: 4 Microservices • 30–40 REST APIs • 6 MongoDB Collections • 20+ React Pages • 5+ Admin Pages • 10+ Internal Packages • 5+ Shared Packages • ~20K–30K LOC

- Architected and built a production-grade e-commerce platform from scratch, migrating from a monolithic backend to a microservices architecture (4 services) behind an API Gateway for improved scalability, fault isolation, and performance.
- Designed a monorepo with 10+ internal and 5+ shared packages (@beautinique/shared-constants, shared-types, shared-zod, backend-constants, backend-types, backend-zod, backend-configs, backend-middlewares, frontend-constants, frontend-types, frontend-zod, frontend-configs) to eliminate duplicate logic and maximize code reuse across services.
- Built a dedicated Media Service to centralize file uploads, deletions, image optimization, and HLS-based video streaming using Cloudinary and Multer.
- Designed secure, multi-provider authentication supporting Google, GitHub, and LinkedIn OAuth alongside JWT, Refresh Tokens, HTTP-only cookies, and role-based access control (RBAC).
- Implemented Redis caching to reduce redundant database queries and BullMQ for asynchronous background job processing, including an email queue system.
- Adopted Zod for scalable, type-safe validation across frontend and backend, paired end-to-end with TypeScript for compile-time safety.
- Built core e-commerce functionality: category and product management, variants, inventory, cart, wishlist, orders, reviews, search, filtering, pagination, and draft products.
- Developed the frontend with React, TypeScript, Tailwind CSS, React Query, Zustand, and React Hook Form; implemented rate limiting and soft-delete patterns on the backend to protect data integrity.

## KEY ENGINEERING HIGHLIGHTS

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- Designed a microservices architecture with an API Gateway and monorepo package structure for a production e-commerce platform.
- Built 10+ reusable internal and shared packages standardizing types, validation, constants, and middleware across services.
- Implemented Redis caching and BullMQ-based asynchronous job processing for scalable backend operations.
- Engineered a secure, multi-provider authentication architecture combining OAuth, JWT, Refresh Tokens, and RBAC.
- Integrated AI/Computer Vision models (TensorFlow.js, MediaPipe) into production-grade real-time virtual try-on features.
- Improved production application responsiveness through rendering optimization and caching strategies.

## ADDITIONAL PROFESSIONAL PROJECTS

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**CMS** — Built CRUD workflows and API integrations using React, TypeScript, React Query, Zustand, React Hook Form, and a Node.js/TypeScript backend.

**CTRUH Website** — Developed the company marketing website using Next.js, TypeScript, and REST APIs.

**PhysicsWallah** — Built data-heavy interfaces and tables with React, TypeScript, React Query, Zustand, and REST API integrations.

**AI Talent POC** — Built React-based data visualization interfaces with graphs and REST API integrations.

**SolarSquare** — Contributed to Solar Host, WhatsApp Flow, and Solar Calculator features.

**Aramco** — Delivered frontend integration work for 3D applications.

## EDUCATION

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**Bachelor of Arts (Economics)** — Yashwantrao Chavan Maharashtra Open University (YCMOU)

2018 – 2021 | 84%

## CERTIFICATION

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**Full Stack Web Development Program** — Masai School

## LANGUAGES

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English, Hindi, Marathi